Saswata Mishra

cs24mtech12001@iith.ac.in | mysaswatamishra@gmail.com | +91 8282950574 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.00
Uttarpara GOVT. High School	Class XII (West Bengal Council Of Higher Secondary Education)	2017	84.40%
Uttarpara Children's Own Home	Class X (West Bengal Board of Secondary Education)	2015	88.85%

PROJECTS

CUDA Warp Specialization & Kernel Optimization | CUDA C++, NVIDIA CUDA, Nsight Compute, Tensor Cores

- Optimized CUDA kernels for **SGEMV, GEMM (FP32/FP64), Tensor Core** GEMM, **3D convolution**, **2D Jacobi** and **stencil computations** using **shared-memory tiling, register tiling, double buffering,** asynchronous copies and warp specialization.
- Developed warp-specialized **producer/consumer kernels** using **cudaDMA** and evaluated alternative GPU optimization strategies across multiple compute kernels on **PolyBenchGPU**.
- Profiled kernel execution with **NVIDIA Nsight Compute**, analyzing memory access, execution and performance characteristics to compare optimization strategies and identify architectural bottlenecks.

TransactiWar -- Secure P2P Money Transfer Application | PHP, MySQL, Docker, Apache, OpenSSL, Python

- Built a **secure peer-to-peer money transfer web application** from scratch with PHP (**no backend framework**), MySQL and Docker, implementing authentication, transaction management, activity logging and secure file uploads.
- Implemented **defence-in-depth security controls** including **prepared statements, CSRF protection, XSS prevention, session hardening, rate limiting, HTTPS, security headers and row-level locking** for concurrent transfers.
- Developed an automated **Python vulnerability scanner** for the adversarial War Game, assessing **17 deployments** of adversary teams for **SQLi, CSRF, XSS, session fixation, file-upload and authentication vulnerabilities**; generated structured reports with evidence.

SM4 Differential Cryptanalysis Suite | Python, NumPy, Cryptography

- Implemented the **32-round SM4 cipher**, key schedule, **S-box, T-function** and **ECB/CBC/OFB/CTR modes** from scratch.
- Generated the S-box Differential Distribution Table (**DDT**) and Linear Approximation Table (**LAT**), and built a **search procedure for 7-round differential trails** following published cryptanalytic results.
- Enumerated and **independently validated** trails with the target activity pattern **without brute-forcing the full 128-bit input space**.

Differential Cryptanalysis of 4-Round DES | Python

- Recovered DES key bits through a chosen-plaintext attack using high-probability differential characteristics and a small number of plaintext-ciphertext pairs.

Local LLM & Vibe-Coding Framework | Docker, Ollama, Open WebUI, NVIDIA CUDA, Linux

- Deployed a self-hosted, GPU-accelerated LLM platform on an NVIDIA RTX A2000 using Docker, Ollama and Open WebUI, enabling authenticated LAN-accessible inference without cloud APIs.
- Designed the deployment lifecycle with automated setup, port selection, model provisioning, GPU health checks, inference smoke tests and teardown/offload scripts.
- Configured GPU passthrough, persistent model/chat storage and isolated service networking, with Ollama restricted to localhost and Open WebUI exposed to the lab LAN.

Breaking RC4 | Python

- Performed a full RC4 key-recovery attack by exploiting statistical biases in the generated keystream.

SKILLS

Languages: Java, Python, C/C++, SQL, Bash, JavaScript;

Cybersecurity & Cryptography: Cryptanalysis, Symmetric Cryptography, Network Security, DNS Security, Web Security, Differential & Linear Cryptanalysis, Quantum Cryptography;

Backend & Systems: Spring Boot, REST APIs, Docker, Kubernetes, Linux, Jenkins, PostgreSQL;

AI/ML: Machine Learning, Agentic AI;

Frameworks & Libraries: CUDA, Scapy, BIND, dnspython, Git, JIRA, Postman, NumPy, pandas, Matplotlib, TensorFlow

POSITIONS OF RESPONSIBILITY

Coordinator -- ACM Summer School on Symmetric Cryptology | IIT Hyderabad

- Coordinated the **10-day ACM Summer School on Symmetric Cryptology**, ensuring planning and execution.
- Also served as a TA for the hands-on in-lab assignment sessions.

Class Representative of NIS Programme

- Elected CR for the NIS programme, batch 2024-25. Guided students regarding academic doubts and issues.

RELEVANT COURSES

Advanced Computer Networks, Network Security, GPU Programming, Cryptology, Topics in Cryptanalysis, Quantum Cryptography (BB84, QKD, QEC), Advanced Topics in Cryptology (ZKP, ZKPoK, Elliptic Curves, Pairings, Lattice Cryptography), Quantum Computing (Shor's & Grover's algorithms, & applications)